Recommender Systems

- Further Approaches –

Video link: https://youtu.be/k9jqWSFpr1k

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1. Conventional Recommendation Approaches

Conventional Recommendation Approaches

Approach	Description	Drawbacks
Collaborative filtering	identify the nearest neighbors in U of the target user u and recommend items in I liked by the nearest neighbors to u	cold-start problems gray-sheep issue quality dependent on historical data set
Content-based filtering	select items in I that are most similar as items liked by the target user u in the past	new-user issue diversity
Demographic-based recommendation	identify users that are demographically most similar as the target user u and recommend items in I liked by these users to u	new-user issue quality dependent on historical data set
Knowledge-based recommendation	recommend items in I that satisfy requirements w.r.t. items of the target user u	the need of knowledge acquisition (e.g., catalog knowledge, functional knowledge, and user knowledge)

2. Hybrid Recommender Systems [BUR2022]

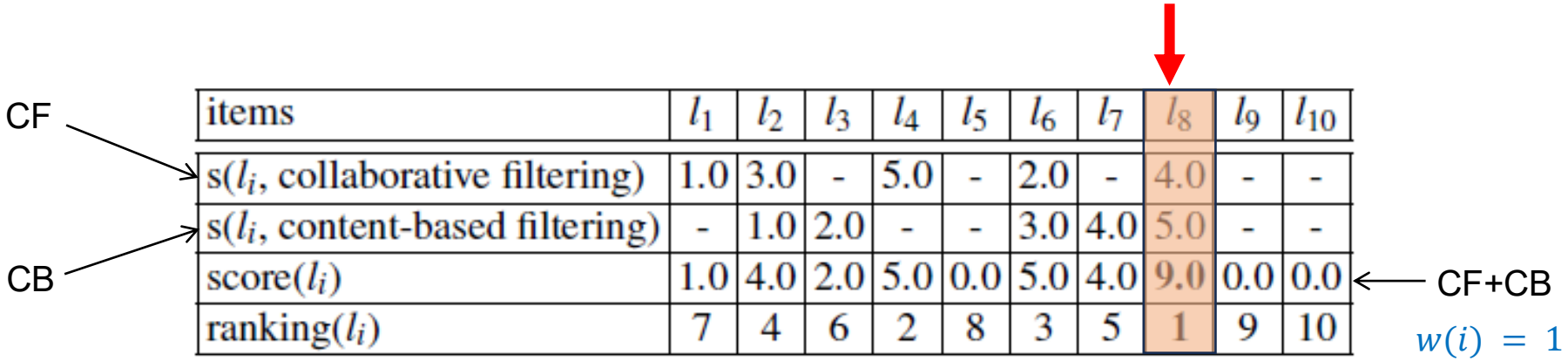
What is a Hybrid Recommender System?

- combines two or more recommendation techniques:
  - gain better performance
  - decrease the drawbacks of any individual one
  - increase prediction accuracy (e.g., if multiple data sources are available)
- Examples:
  - combining *collaborative filtering* with *content-based filtering* (e.g., the *DailyLearner* system [BIL2000])
  - combining *collaborative filtering* with *knowledge-based recommendation* (e.g., the *FindMe* system [BUR2000])

Hybrid Recommendation Approaches [BUR2002]

Hybrid rec. approach	Description
Weighted	The scores of several recommendation techniques are combined to produce a single recommendation.
Mixed	Recommendations from several recommenders are presented at the same time.
Cascade	One recommender refines the recommendations given by another.
Switching	The system switches between recommendation techniques depending on the current situation.
Feature combination	Features from different recommendation data sources are thrown together into a single recommendation algorithm.
Feature augmentation	Output from one recommendation technique is used as an input to another.
Meta-level	The model learned by one recommender is used as input to another.

Weighted Hybrid Recommendation

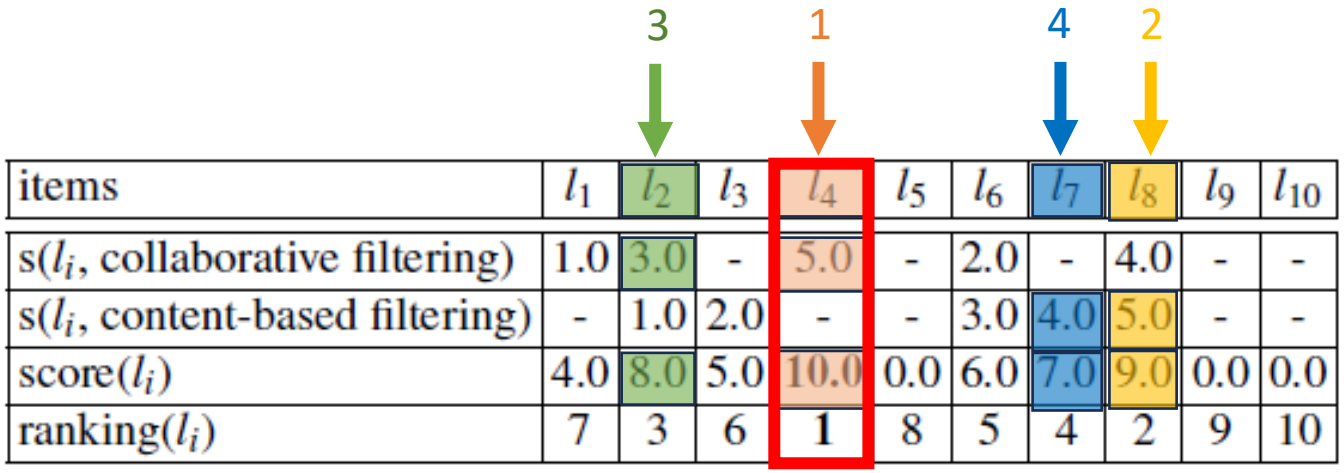


items	l_1	l_2	l_3	l_4	l_5	l_6	l_7	l_8	l_9	l_{10}
s(l_i, collaborative filtering)	1.0	3.0	-	5.0	-	2.0	-	4.0	-	-
s(l_i, content-based filtering)	-	1.0	2.0	-	-	3.0	4.0	5.0	-	-
score(l_i)	1.0	4.0	2.0	5.0	0.0	5.0	4.0	9.0	0.0	0.0
ranking(l_i)	7	4	6	2	8	3	5	1	9	10

Table 11: Example of *weighted* hybrid recommendation: individual predictions are integrated into one score. Item *l*8 receives the best overall score (9.0).

$$score(item) = \sum_{rec \in RECS} s(item, rec) * w(i)$$

Mixed Hybrid Recommendation



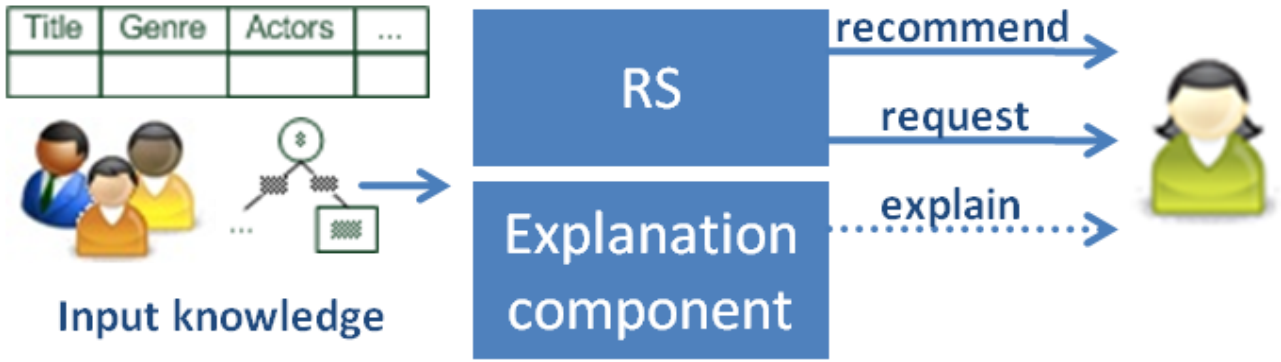
items	l_1	l_2	l_3	l_4	l_5	l_6	l_7	l_8	l_9	l_{10}
s(l_i, collaborative filtering)	1.0	3.0	-	5.0	-	2.0	-	4.0	-	-
s(l_i, content-based filtering)	-	1.0	2.0	-	-	3.0	4.0	5.0	-	-
score(l_i)	4.0	8.0	5.0	10.0	0.0	6.0	7.0	9.0	0.0	0.0
ranking(l_i)	7	3	6	**1**	8	5	4	2	9	10

Table 12: Example of *mixed* hybrid recommendation: individual predictions are integrated into one score conform the zipper principle (best collaborative filtering prediction receives score=10.0, best content-based filtering prediction receives score=9.0 and so forth).

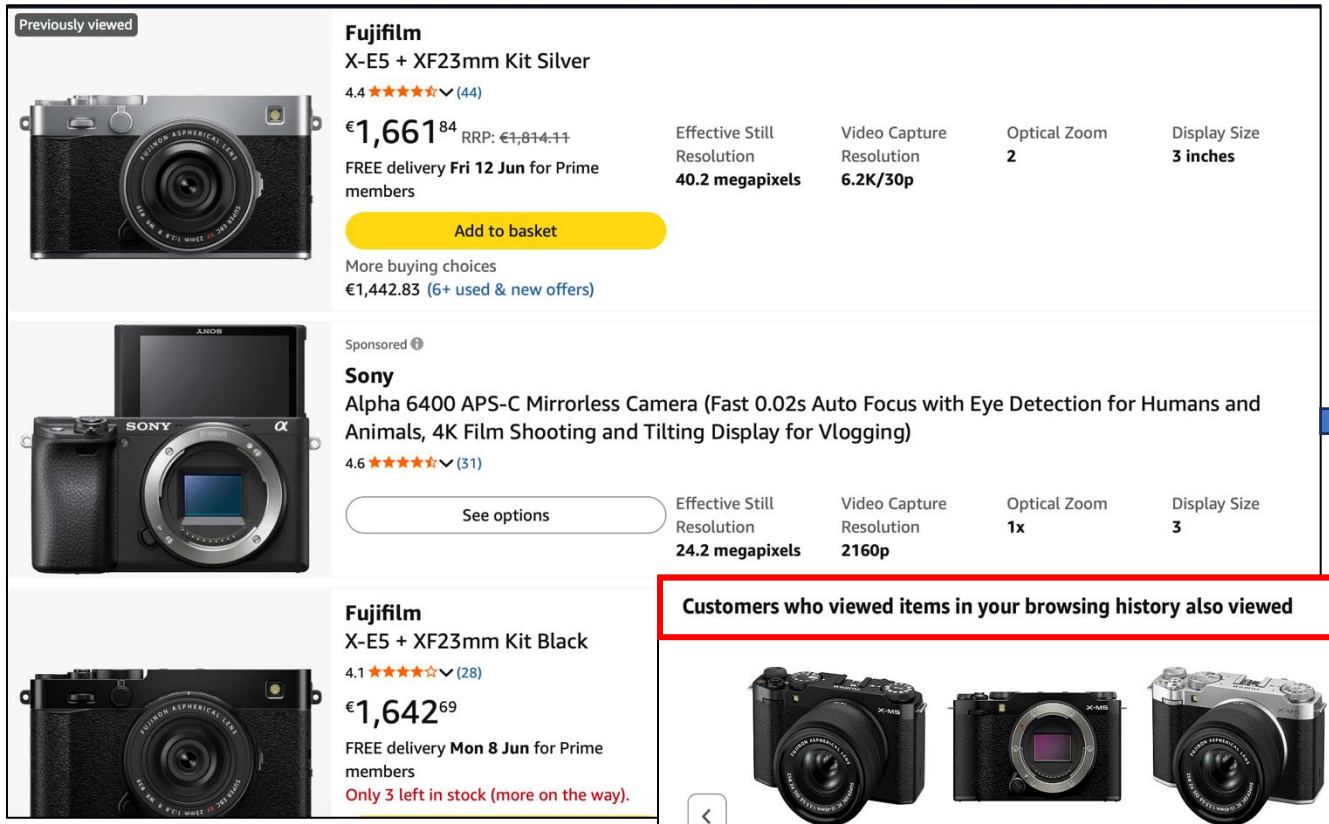
3. Explanations in Recommender Systems
[FEL2018B, TIN2007, TRA2019, TRA2023, TRA2024]

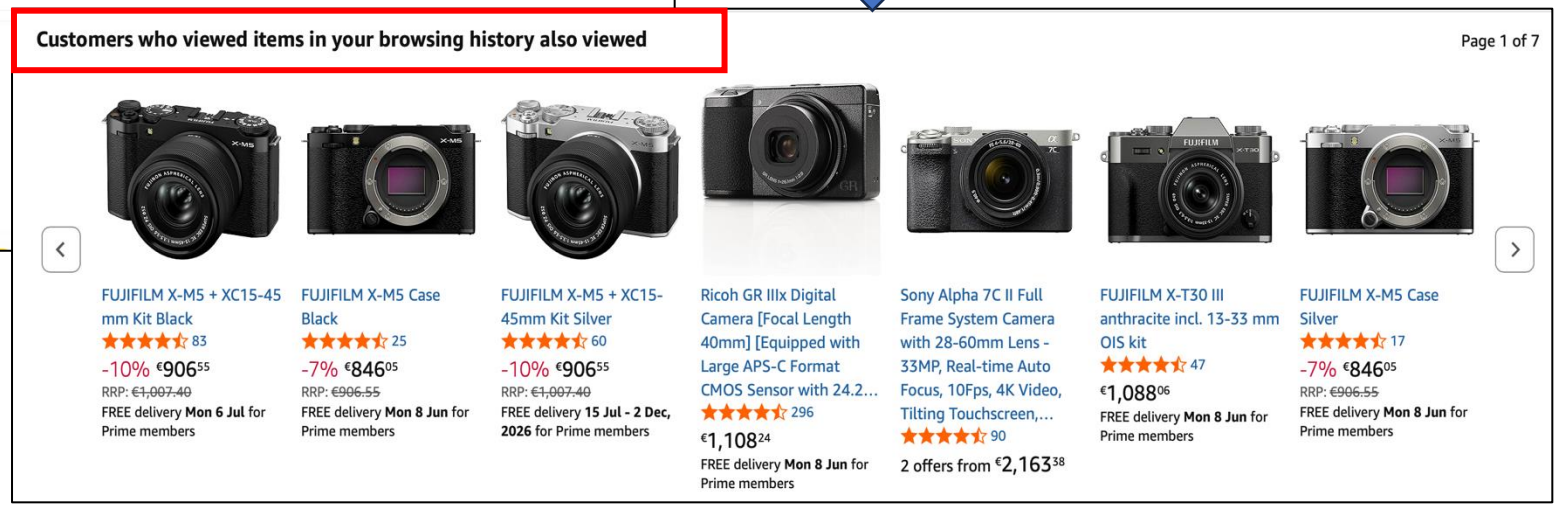
What is an "Explanation" in Recommender Systems?

Additional information to explain the system's output following explanation goals

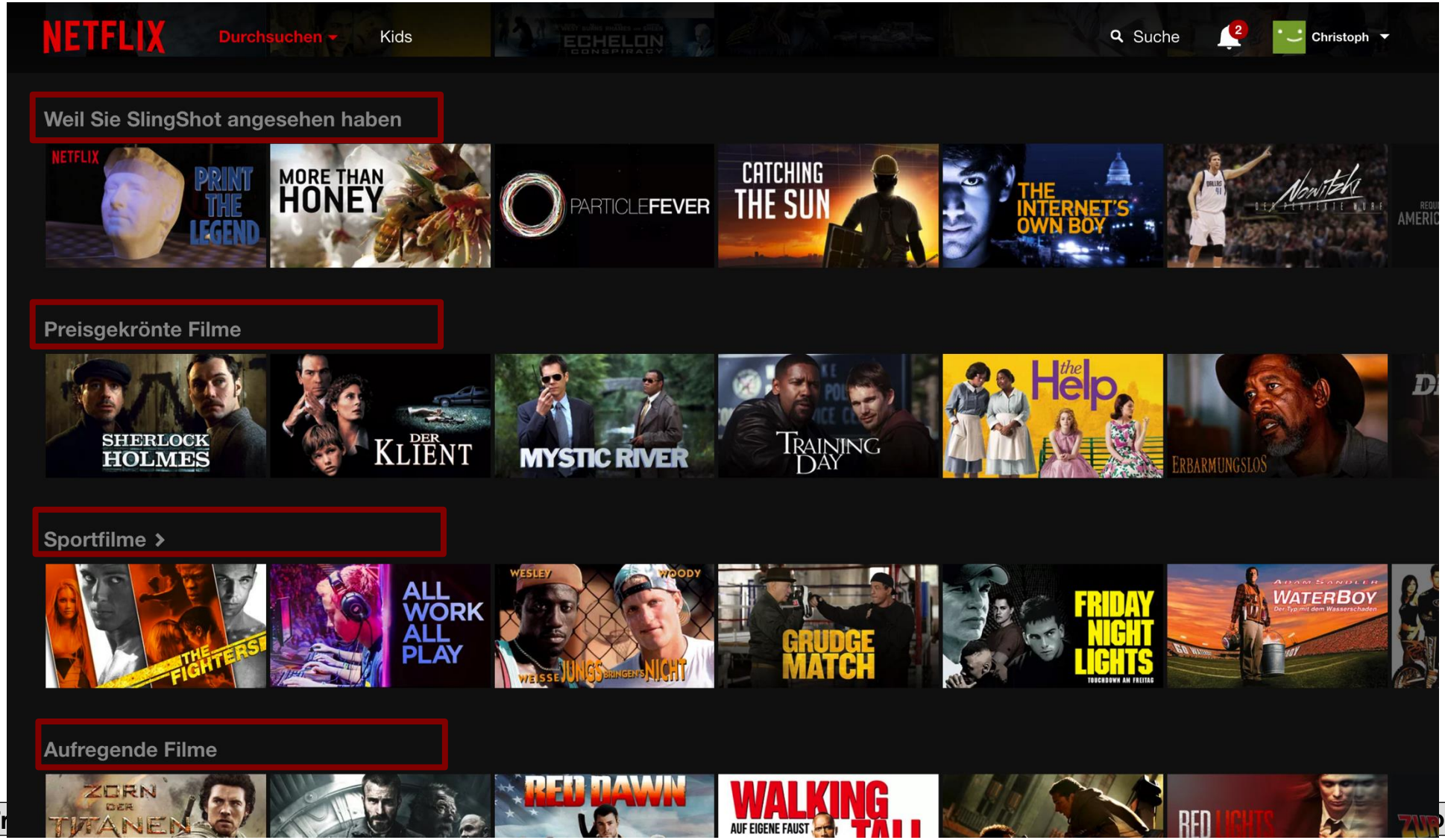


Explanations in Amazon

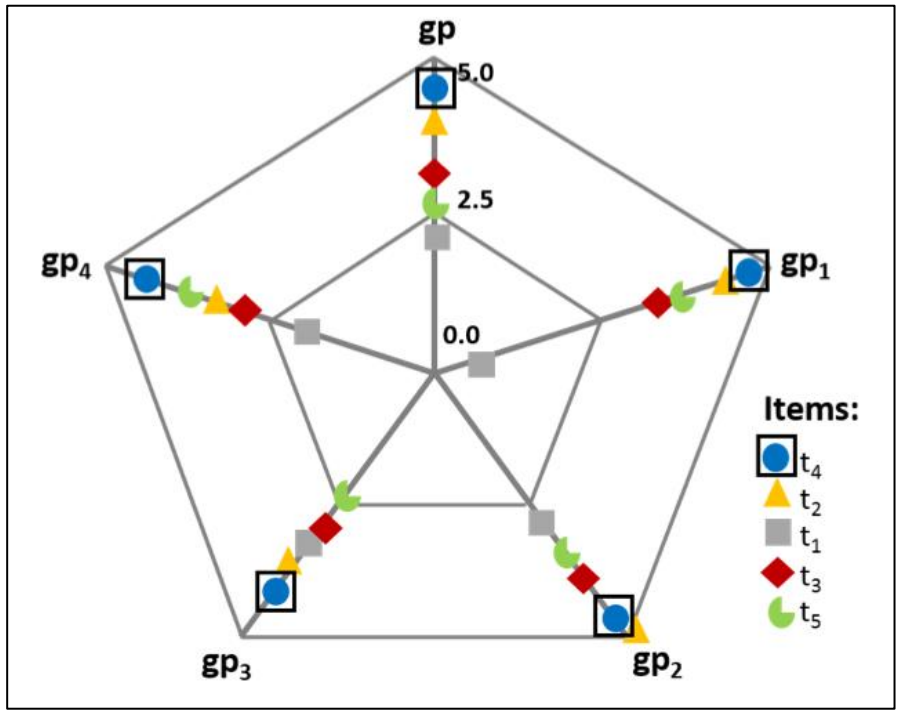




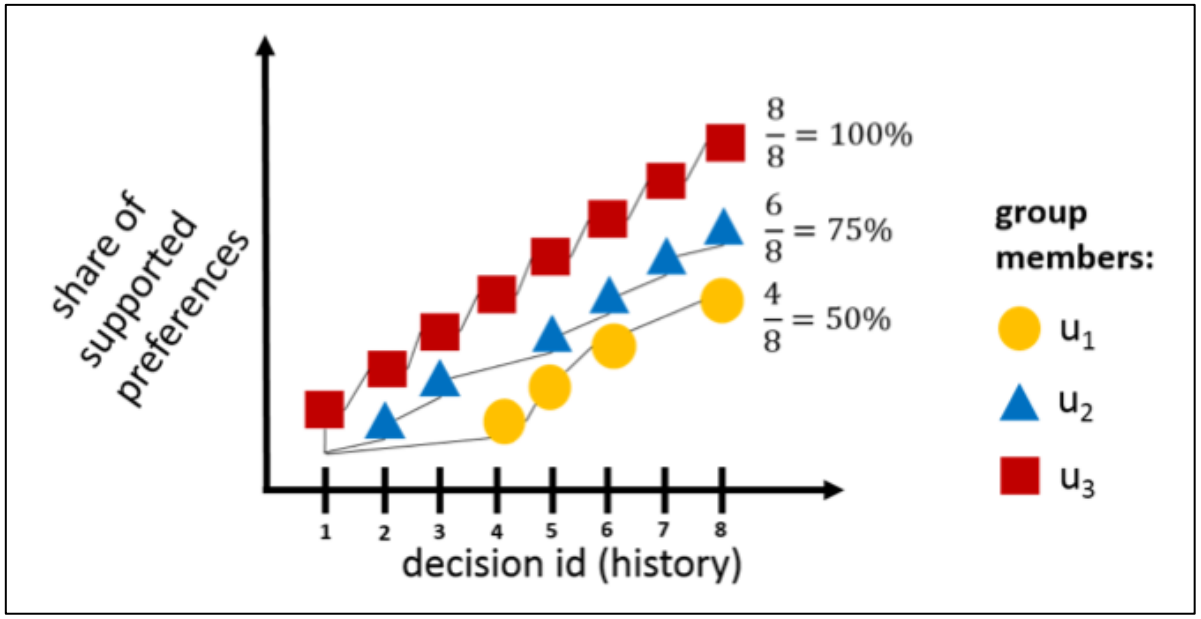
Explanations in Netflix



Explanations of group recommendations



Explaining why item t_4 is recommended to the group **[FEL2018B]**.



Visualizing the *degree of fairness* in repeated group decisions. Group member u_1 was at a disadvantage in previous decisions **[FEL2018B]**.

Goals: Why Provide Explanations?

Goals of Explanations (1)

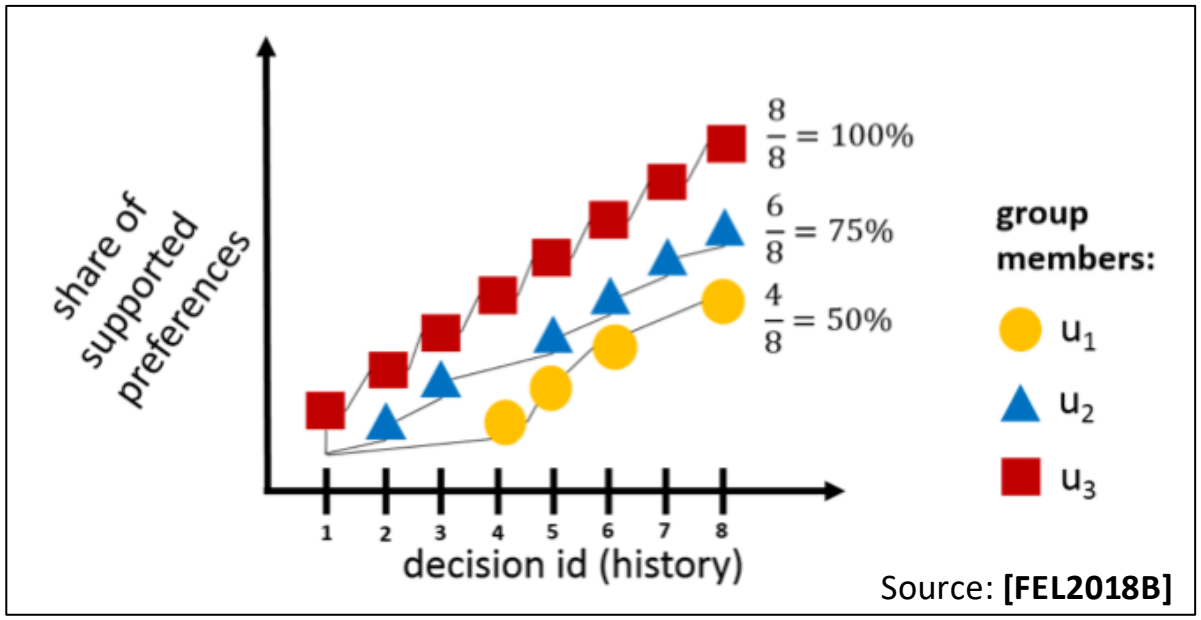
- **Transparency**: provide the basis to understand *how the system works* (e.g., in terms of *how*? explanations)
- **Interpretability**: help users to understand the system output (e.g., in terms of *why*? explanations)
- **Scrutability**: allow users *to tell the system it is wrong* (e.g., wrong assumptions about user preferences)
- **Trust**: *increase users' confidence* in the system
- **Effectiveness**: *help users make good decisions*
- **Persuasiveness**: *convince users* to try or buy
- **Efficiency**: *help users to make decisions faster*

Goals of Explanations (2)

- **Satisfaction**: *increase usability/enjoyment and output quality*
- **Fairness**: *understanding/avoiding biases* in recommender systems
- **Accountability**: *help users to hold recommender systems accountable*
- **Relevance:** *justify why additional information is needed from the user*
  - In financial services recommendations, further information regarding *risk level* and *fund type* are needed.
  - *Example explanation: "We could provide you with a better recommendation if we know which risk level you are willing to take and what kind of funds you want to have".*
- **Education:** *help them better understand the product domain*
  - Deep knowledge about the domain helps customers rethink their preferences and evaluate the pros and cons of different solutions.
  - Eventually, as customers become more informed, they can make wiser purchasing decisions.

Further Goals of Explanations in Group Recommender Systems (1)

- **Fairness-related explanations:**
  - clarify how well the preferences of group members are considered
  - increase fairness perception of group members



Source: **[FEL2018B]**

> *"Item A has been recommended to the group since no group member has a real problem with it. Besides, this decision supports the preferences of users u_1 who were treated less favorably in the last decisions."* **[TRA2019]**

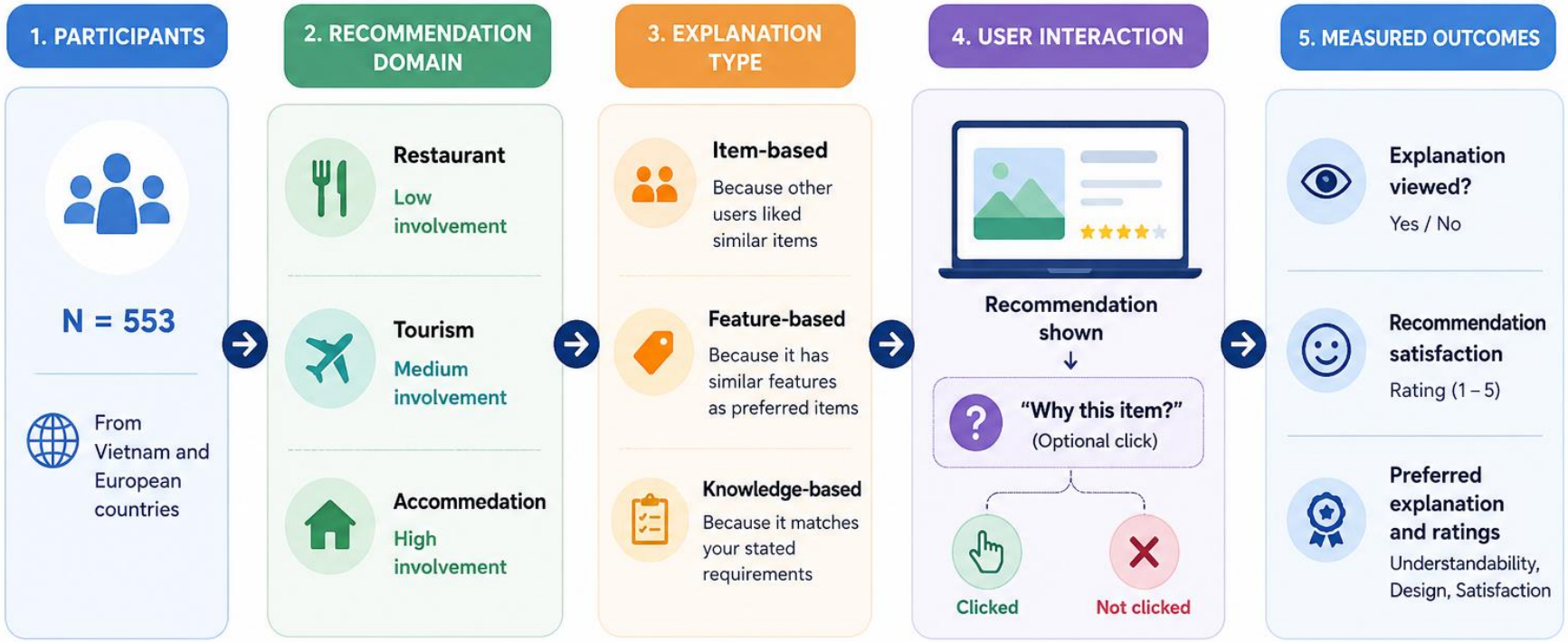
When Provide Explanations?

When do users need explanations? [TRA2023]

- Explainable recommender systems focuses on:
  - *How to generate explanations?*
  - *How explanations improve explanation goals?*
- However, a more fundamental question remains:
  - ***Do users actually want explanations, and when?***

USER STUDY DESIGN

Investigating when users need explanations, across domains and explanation types



KEY IDEA

Users were free to decide whether to look at explanations.
We analyze **natural explanation-seeking behavior.**

MAIN GOALS

- When do users need explanations?
- Does need depend on domain?
- Do personal characteristics matter?
- Which explanation type is preferred?

Source: **[TRA2023]**

Key Findings [TRA2023]

- **Users do not always want explanations:** Explanation needs depend on the decision context.
- **High-stakes decisions (e.g., accomodation):** Explanations are needed when recommendations are **unsatisfactory**.
- **Low-/average-stakes decisions (e.g., songs and travel destinations):** Explanations are often viewed when recommendations are **satisfactory**.

Persuasive Explanations

Persuasion Principles [CIA2001]

persuasion principle	semantics
reciprocity	feeling of an obligation to give something back
scarcity	reduced item availability increases preparedness to purchase
authority	experts have an increased influence over users
commitment	users prefer to be consistent with their articulated preferences
liking	users like other users who are similar to themselves
social proof	users follow the opinions (of a representative set) of other users

emphasize positive contributions or shared benefits

highlight reduced availability or limited opportunities

show scientific studies, or opinions /evaluations of experts or well-known organizations.

remind users of their earlier intentions

refer to users with similar interests or lifestyles

show that many users selected the recommended item and achieve good results

Sustainability-Aware Explanations [TRA2024]

principle	explanation
reciprocity	Think about purchasing a smaller car, i.e., not an SUV. With this, you will contribute to sustainable cities of the future and also help to make life worth living for the upcoming generations.
scarcity	Purchasing an SUV is worth re-thinking, the longer we follow this line of mobility philosophy, the less opportunities we will have to achieve the goal of sustainable cities and communities.
authority	Environmental scientists have emphasized recently that the overall share of SUVs significantly increases which results in an increased C02 emission and increasing parking infrastructure issues. Think about a smaller car since your goal is just to be mobile in the local environment.
commitment	Since you declared your intention to contribute to a sustainable city of the future which includes the aspects of reduced CO2 emission and also efficient handling of parking infrastructures, you should think about purchasing a smaller car instead of an SUV.
liking	Customers interested in sustainable cities and communities finally decided to not purchase an SUV but to purchase a smaller car instead.
social proof	Most customers searching for SUVs who just want to be mobile in the local environment often decided to switch from a larger to smaller cars.

4. Evaluating Recommender Systems

[EVA2022, FEL2018A, SHA2011]

Ways to Evaluate a Recommender System

- **Online evaluation:**
  - lets users interact with a real system
  - sees how well the recommended items match the users/participants' interests.
  - "*Between subjects*": each subject is assigned to exactly one condition.
  - "Within subjects": each subject is assigned to a set of conditions.
  - PROS: most realistic
  - CONS: time intense, costly
- **Offline evaluation:**
  - Primarily based on datasets that are used for evaluation purposes
  - PROS: very cheap, no users involved
  - CONS: often not very accurate

Online Evaluation

Example: Online Evaluation (1)

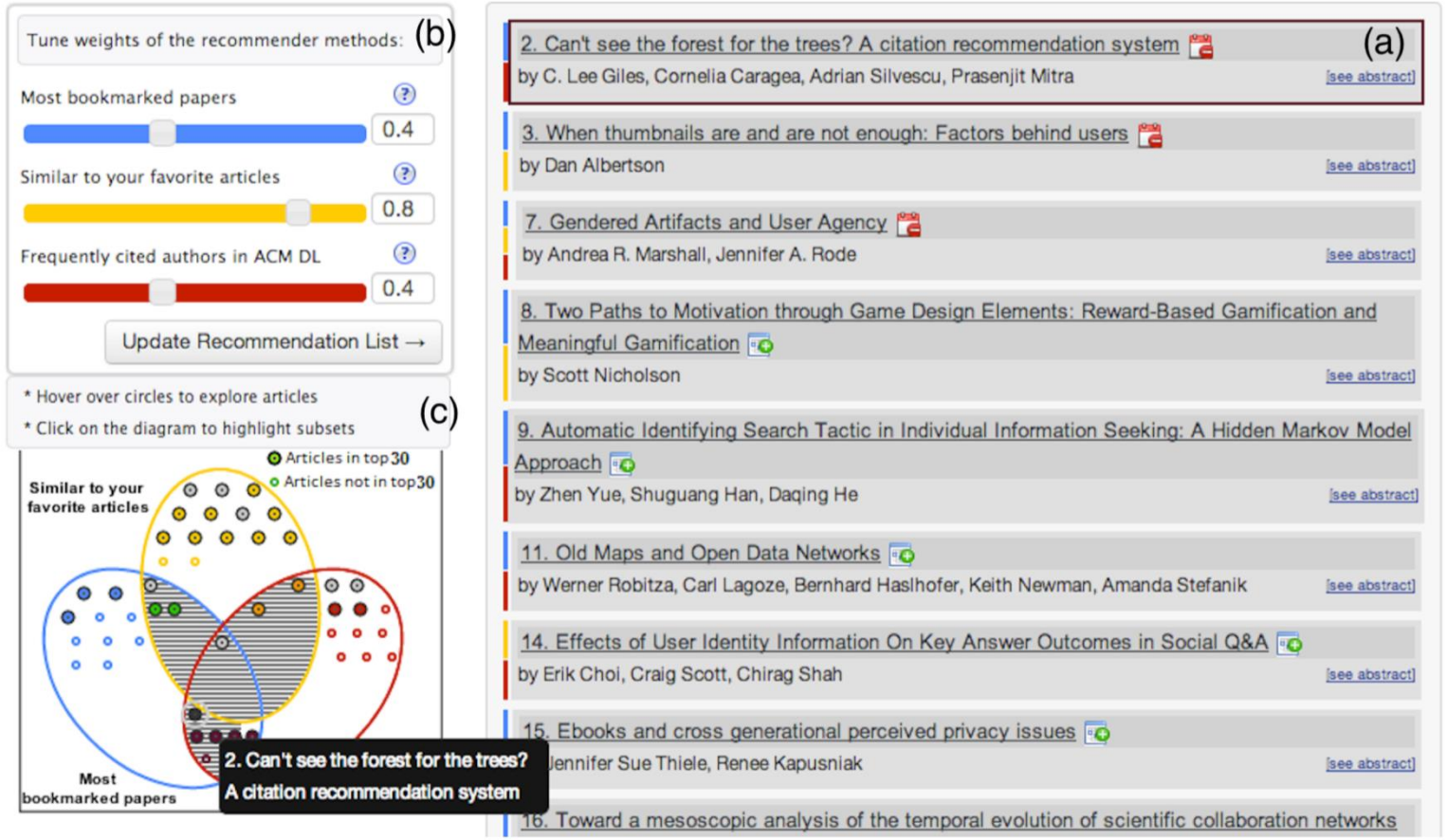
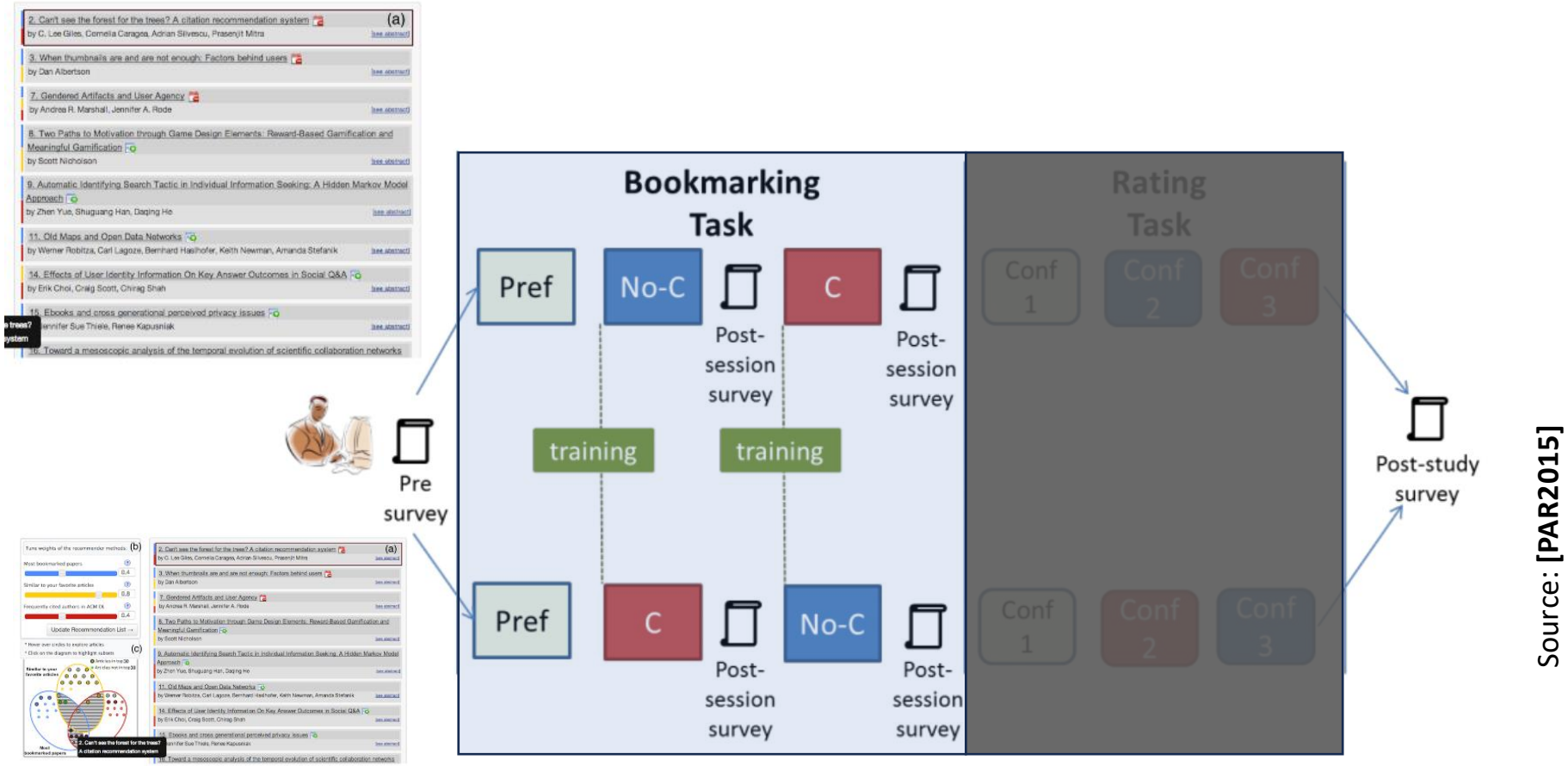


Figure 1. Screenshot of SetFusion displaying (a) a filtered list of papers recommended, (b) sliders, and (c) the Venn diagram.

Source: Para et al. **[PAR2015]**

Example: Online Evaluation (2)



Source: [PAR2015]

Figure 5. Workflow of user study. After answering the entry questionnaire (pre-survey), the subject was assigned to one of 2 possible sequences of interfaces: Non-controllable (No-C) and then controllable (C) interface, or vice versa.

Issue with Online Evaluation

Not always clear differences can be found **[PAR2015]**

"In our within-subjects study with 40 participants, the average rating was 2.48 ±0.089, while the controllable interface had an average rating of 2.46 ±0.076. On that general level, ***there were no significant differences between the interfaces****."*

Offline Evaluation (Simulations)

Offline Evaluations

- **Main idea:** perform evaluation on historical data (data sets)
- **Split data** in training set and test set
- **Train the algorithm** on the training set and validate on the test set
- **Use a set of evaluation metrics** to investigate the extend to which the recommender method performs

Evaluation Metrics

Classification Metrics (1)

- Recommendation problem
  - considered *information retrieval task*
  - recommends items which are classified to be “good”.
- Typical classification metrics:
  - **Precision** measures the exactness of the recommendation: *the proportion of relevant items among the recommended items*.
    - E.g., the proportion of recommended movies that are regarded as “good”.

$$precision = \frac{|\{relevant\ doccuments\} \cap \{retrieved\ documents\}|}{|\{retrieved\ documents\}|}$$

  - **Recall** measures the completeness of the recommendation: *the proportion of relevant items that were successfully recommended out of the total number of relevant items in the dataset*.
    - E.g., the proportion of all good movies recommended.

$$recall = \frac{|\{relevant\ documents\} \cap \{retrieved\ documents\}|}{|\{relevant\ documents\}|}$$

Classification Metrics (2)

- The F_1 metric:
  - attempts to combine precision and recall into a single metric
  - provides a balanced evaluation of the recommender system's performance

$$F_1 = 2.\frac{precision \cdot recall}{precision + recall}$$

Error Metrics

- Error metrics measure error rate.
- **Underlying assumption**: The small the error, the better the algorithm.
- Typical error metrics:
  - **Mean Absolute Error** (*MAE*) computes the deviation between predicted ratings and actual ratings.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|p_i - r_i|$$

  - **Root Mean Square Error** (*RMSE*) is similar to *MAE,* but *emphasizes more on larger deviation.*

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(p_i - r_i)^2}$$

Ranking Metrics

- Ranking metrics are used to evaluate the quality of ranked recommendations.
- Typical ranking metrics:
  - **Normalized Discounted Cumulative Gain** (*NDCG*):
    - considers both relevance and position of recommended items
    - items ranked higher have higher scores
    - items ranked lower have discounted scores
  - **Mean Reciprocal Rank** (*MRR*):
    - focuses on the position of the first relevant item in the recommendation list

Novelty and Serendipity Metrics

- **Novelty metrics:**
  - measure to which extent recommended items are novel
  - refer to the fraction of recommended items that are indeed new to the user
- **Serendipity metrics**:
  - tightly related to novelty
  - measure how surprising recommendations are to a user
  - can be measured by the fraction of relevant and unexpected recommendations in the list

For more details of the mentioned evaluation metrics, see the studies of Eva Zangerle et al. **[EVA2022]** and Shani et al. **[SHA2011]**.

5. Exercises

Exercise 1

- Develop your own example of a **mixed hybrid** recommendation (10 items and 2 recommenders)
- Develop your own example of a **weighted hybrid** recommendation (10 items and 2 recommenders)
- **Note**: no submission („Abgabe“) needed!

Exercise 2

- Perform an Internet search to find one example of a recommender explanation interface – show and explain this example w.r.t. potential related goals
- **Note**: no submission („Abgabe“) needed!

Exercise 3

- Determine MAE and RMSE for the following rating information:
- Rating data (actual ratings):

	User1	User2	User3	User4	User5
Item 1	3.0	3.0	4.0	3.0	2.0
Item 2	2.0	1.0	3.0	2.0	2.0
Item 3	4.0	3.0	5.0	4.0	4.0

- Predicted ratings:

	User1	User2	User3	User4	User5
Item 1	2.0	2.0	3.0	4.0	2.0
Item 2	2.0	2.0	2.0	3.0	2.0
Item 3	4.0	5.0	4.0	4.0	3.0

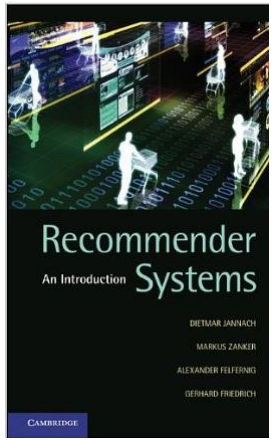
- **Note**: no submission („Abgabe“) needed!

Exercise 4

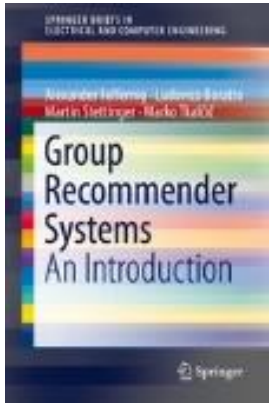
- Perform an Internet search to find other examples of recommender frameworks (at least one).
- **Note**: no submission („Abgabe“) needed!

Course Material

- D. Jannach, M. Zanker, A. Felfernig, and G. Friedrich. Recommender Systems – An Introduction, Cambridge University Press, 2010 (*can be found in library*).



- A. Felfernig, M. Stettinger, L. Boratto, and M. Tkalcic. Group Recommender Systems – An Introduction, Springer, 2018.



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Thank you for your attention!